

Light chaser

Test cases:

Test case	Stimulus	Expected behavior	Result
Free-run shift	reset_n de-asserted, hold_n=1	One-hot bit shifts once per clk_out period, full 10-bit wraparound	Pass
Hold	hold_n=0 for 3 clk_out periods	shift_out remains constant	Pass
Resume after hold	hold_n returns to 1	Shifting resumes from last state	Pass
Async reset mid-shift	reset_n=0 pulsed mid-sequence	shift_out immediately returns to initial state, independent of clock edge	Pass
Clock division accuracy	Measured clk_out period vs. calculated	125 ms measured vs. 125 ms calculated (50 MHz/6.25M divisor)	Match

Where, $187,500,010,000 - 62,500,010,000 = 125,000,000,000 \text{ ns}$

$125,000,000,000 \text{ ns} \div 1,000,000 = 125,000 \text{ ms}$

$\text{DIVISOR} \times T_{\text{clk_in period}} = 6,250,000 \times 20 \text{ ns} = 125,000,000 \text{ ns} = 125 \text{ ms}$

Result:

All test cases produced outputs matching the expected values:

```

# t=0 reset_n=0 hold_n=1 shift_out=1000000000
# t=12000 reset_n=1 hold_n=1 shift_out=1000000000
# t=62500010000 reset_n=1 hold_n=1 shift_out=0100000000
# t=187500010000 reset_n=1 hold_n=1 shift_out=0010000000
# t=312500010000 reset_n=1 hold_n=1 shift_out=0001000000
# t=437500010000 reset_n=1 hold_n=1 shift_out=0000100000
# t=562500010000 reset_n=1 hold_n=1 shift_out=0000010000
# t=687500010000 reset_n=1 hold_n=1 shift_out=0000001000
# t=812500010000 reset_n=1 hold_n=1 shift_out=0000000100
# t=937500010000 reset_n=1 hold_n=1 shift_out=0000000010
# t=1062500010000 reset_n=1 hold_n=1 shift_out=0000000001
# t=1187500010000 reset_n=1 hold_n=1 shift_out=1000000000
# t=1312500010000 reset_n=1 hold_n=1 shift_out=0100000000
# t=1375000010000 reset_n=1 hold_n=0 shift_out=0100000000
# t=1750000010000 reset_n=1 hold_n=1 shift_out=0100000000
# t=1812500010000 reset_n=1 hold_n=1 shift_out=0010000000
# t=1937500010000 reset_n=1 hold_n=1 shift_out=0001000000
# t=2062500010000 reset_n=1 hold_n=1 shift_out=0000100000
# t=2187500010000 reset_n=1 hold_n=1 shift_out=0000010000
# t=2250000013000 reset_n=0 hold_n=1 shift_out=1000000000
# t=2250000020000 reset_n=1 hold_n=1 shift_out=1000000000
# t=2312500010000 reset_n=1 hold_n=1 shift_out=0100000000
# t=2437500010000 reset_n=1 hold_n=1 shift_out=0010000000
# t=2562500010000 reset_n=1 hold_n=1 shift_out=0001000000
# t=2687500010000 reset_n=1 hold_n=1 shift_out=0000100000
# ** Note: $finish : C:/Users/hp/Desktop/Digital design assignments/task2/tb.v(57)
# Time: 2750000010 ns Iteration: 1 Instance: /tb light chaser
  
```

